M3-T02: tumor change between two scans of the same patient

Volume change in percent, `(V_later - V_earlier) / V_earlier x 100`, per region. Volumes = voxels x voxel volume (cm3); regions follow the team's REGIONS (TC = labels 1+3, WT = 1+2+3, ET = 3). 'expert' = volumes from the expert masks (reference); 'nnU-Net' = volumes from seg-model v1 predictions on the same scans.

RHUH-0002

Interval	Between	Region	Volume before (cm3)	Volume after (cm3)	Change (expert)	Change (nnU-Net)
pre-op -> early post-op	about 9 days	TC	1.60	0.00	-100.0%	+1038.1%
pre-op -> early post-op	about 9 days	WT	28.66	6.64	-76.8%	+47.4%
pre-op -> early post-op	about 9 days	ET	1.25	0.00	-100.0%	+226.5%
early post-op -> recurrence	interval not stated	TC	0.00	4.60	n/a	-66.6%
early post-op -> recurrence	interval not stated	WT	6.64	10.49	+58.0%	-51.6%
early post-op -> recurrence	interval not stated	ET	0.00	4.29	n/a	-35.0%
pre-op -> recurrence	interval not stated	TC	1.60	4.60	+187.6%	+280.6%
pre-op -> recurrence	interval not stated	WT	28.66	10.49	-63.4%	-28.6%
pre-op -> recurrence	interval not stated	ET	1.25	4.29	+242.1%	+112.1%

- pre-op -> early post-op: the whole tumor (WT) shrank by 76.8% (28.7 to 6.6 cm3, expert; nnU-Net says +47.4%).
- early post-op -> recurrence: the whole tumor (WT) grew by 58.0% (6.6 to 10.5 cm3, expert; nnU-Net says -51.6%).
- pre-op -> recurrence: the whole tumor (WT) shrank by 63.4% (28.7 to 10.5 cm3, expert; nnU-Net says -28.6%).

RHUH-0003

Interval	Between	Region	Volume before (cm3)	Volume after (cm3)	Change (expert)	Change (nnU-Net)
pre-op -> early post-op	about 6 days	TC	15.26	0.00	-100.0%	+61.8%
pre-op -> early post-op	about 6 days	WT	41.43	1.05	-97.5%	+22.6%
pre-op -> early post-op	about 6 days	ET	14.02	0.00	-100.0%	-74.3%
early post-op -> recurrence	interval not stated	TC	0.00	5.07	n/a	-82.2%
early post-op -> recurrence	interval not stated	WT	1.05	37.37	+3469.1%	-25.1%
early post-op -> recurrence	interval not stated	ET	0.00	4.60	n/a	-2.3%
pre-op -> recurrence	interval not stated	TC	15.26	5.07	-66.8%	-71.1%
pre-op -> recurrence	interval not stated	WT	41.43	37.37	-9.8%	-8.1%
pre-op -> recurrence	interval not stated	ET	14.02	4.60	-67.2%	-74.9%

- pre-op -> early post-op: the whole tumor (WT) shrank by 97.5% (41.4 to 1.0 cm3, expert; nnU-Net says +22.6%).
- early post-op -> recurrence: the whole tumor (WT) grew by 3469.1% (1.0 to 37.4 cm3, expert; nnU-Net says -25.1%).
- pre-op -> recurrence: the whole tumor (WT) shrank by 9.8% (41.4 to 37.4 cm3, expert; nnU-Net says -8.1%).

RHUH-0004

Interval	Between	Region	Volume before (cm3)	Volume after (cm3)	Change (expert)	Change (nnU-Net)
pre-op -> early post-op	about 5 days	TC	13.14	0.00	-100.0%	+93.3%
pre-op -> early post-op	about 5 days	WT	118.41	45.38	-61.7%	-26.6%
pre-op -> early post-op	about 5 days	ET	8.69	0.00	-100.0%	-85.5%
early post-op -> recurrence	interval not stated	TC	0.00	37.41	n/a	+33.1%
early post-op -> recurrence	interval not stated	WT	45.38	217.67	+379.6%	+156.7%
early post-op -> recurrence	interval not stated	ET	0.00	35.74	n/a	+2341.2%

Interval	Between	Region	Volume before (cm3)	Volume after (cm3)	Change (expert)	Change (nnU-Net)
pre-op -> recurrence	interval not stated	TC	13.14	37.41	+184.7%	+157.2%
pre-op -> recurrence	interval not stated	WT	118.41	217.67	+83.8%	+88.5%
pre-op -> recurrence	interval not stated	ET	8.69	35.74	+311.6%	+254.8%

- pre-op -> early post-op: the whole tumor (WT) shrank by 61.7% (118.4 to 45.4 cm3, expert; nnU-Net says -26.6%).
- early post-op -> recurrence: the whole tumor (WT) grew by 379.6% (45.4 to 217.7 cm3, expert; nnU-Net says +156.7%).
- pre-op -> recurrence: the whole tumor (WT) grew by 83.8% (118.4 to 217.7 cm3, expert; nnU-Net says +88.5%).

How well nnU-Net matches the expert masks

Volumes per timepoint (correlation between nnU-Net and expert volumes over the patients, medians in cm3):

timepoint	region	correlation	expert_median_cm3	nnunet_median_cm3	median_abs_error_cm3
pre-op	TC	1.00	31.18	30.08	0.86
pre-op	WT	1.00	99.48	105.83	1.69
pre-op	ET	1.00	17.39	16.37	0.78
early post-op	TC	0.03	0.00	32.19	32.19
early post-op	WT	0.80	24.99	84.15	57.46
early post-op	ET	0.36	0.00	2.13	2.13
recurrence	TC	0.58	9.66	24.80	1.98
recurrence	WT	0.89	44.13	83.06	11.00
recurrence	ET	0.54	8.74	11.08	1.49

% change per interval and region (n = patients where the % is defined in both; same_direction = share of patients where nnU-Net gets grew/shrank right; median_abs_diff_points = median difference in percentage points):

pair	region	n	same_direction	median_abs_diff_points
pre-op -> early post-op	TC	27	0.44	107.39
pre-op -> early post-op	WT	27	0.74	52.61
pre-op -> early post-op	ET	27	0.93	14.89
early post-op -> recurrence	WT	27	0.70	124.52
pre-op -> recurrence	TC	27	0.81	27.41
pre-op -> recurrence	WT	27	0.89	14.09
pre-op -> recurrence	ET	27	0.93	13.85
early post-op -> recurrence	TC	9	0.56	4363.97
early post-op -> recurrence	ET	9	1.00	3155.77

- Best interval for the automatic method: pre-op -> recurrence (direction right in 88%, median difference 16.9 points over all regions).
- Weakest interval: pre-op -> early post-op (direction right in 70%, median difference 51.0 points).
- On the pre-operative scans, the kind the model was trained on, nnU-Net and expert volumes correlate at r >= 1.00 (median error up to 1.7 cm3), so the volume measurement itself is sound.
- Our expert volumes reproduce the dataset's own clinical volumes (enhancing tumor and T2/FLAIR abnormality, before and after surgery) with a largest difference of 5.40 cm3 over 27 patients.

All 27 patients: median change

Median % change per interval and region, over the patients where the percentage is defined for both the expert masks and nnU-Net.

Interval	Region	Patients (n)	Median change (expert)	Median change (nnU-Net)
pre-op -> early post-op	TC	27	-100.0%	+7.4%

Interval	Region	Patients (n)	Median change (expert)	Median change (nnU-Net)
pre-op -> early post-op	WT	27	-73.9%	-16.0%
pre-op -> early post-op	ET	27	-100.0%	-85.1%
early post-op -> recurrence	WT	27	+145.6%	+6.6%
pre-op -> recurrence	TC	27	-55.1%	+22.3%
pre-op -> recurrence	WT	27	-23.4%	-6.6%
pre-op -> recurrence	ET	27	-9.8%	+22.2%
early post-op -> recurrence	TC	9	+4351.7%	+25.1%
early post-op -> recurrence	ET	9	+3565.6%	+1060.8%

Notes and limits

- '% n/a' means the region was smaller than 0.1 cm3 in the first scan, so a percentage would be meaningless; use the change in cm3 instead.
- The model was trained on pre-treatment BraTS 2021 scans. Post-operative and recurrence scans are outside that range, so the expert-mask numbers are the reference and the nnU-Net numbers are a check on the method.
- RHUH-GBM labels: 1 = necrosis, 2 = peritumoral signal (edema plus non-enhancing tumor), 3 = enhancing tumor. TC (labels 1+3) leaves out non-enhancing tumor, so rely on WT and ET.
- Patients: RHUH-GBM has 40 patients; 27 have all 3 timepoints in the cleaned data. The others are left out because cleaned scans are missing (7 with incomplete folders, the rest have no cleaned folder).
- Intervals: for pre-op -> early post-op the days are the days from the earliest imaging to surgery in the clinical file (the early post-op scan is within 72 h of surgery, so the real gap can be up to 3 days longer). The clinical file has no recurrence-scan date, so the other intervals are not stated.
- Intervals that include the early post-operative scan agree least, which fits with the model never seeing post-operative scans in training (surgical cavity, blood products). Use the expert-based percentages as the reference and treat the nnU-Net-based ones as an estimate that is only reliable where the tables above say so.
- Early post-operative scans can show surgical changes (cavity, blood products) that no region label describes.
- Scan source: cleaned. Days between scans come from the dataset's clinical data only when given in the config.

- Percent changes of small regions swing widely: a few voxels of difference can look like a large percentage.